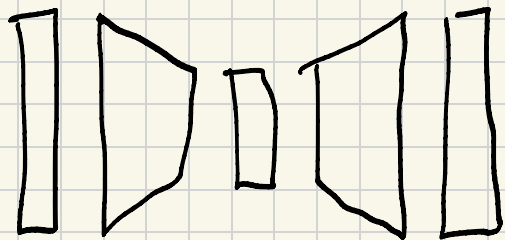


Tuesday, April 7

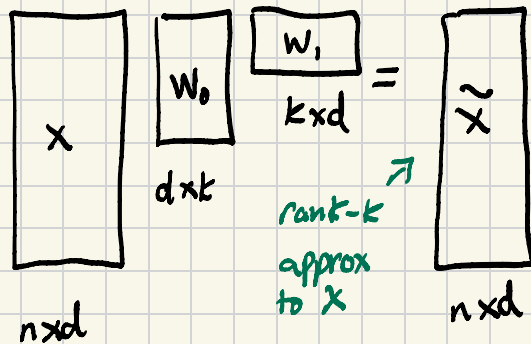
- Eckart-Young-Mirsky Theorem

Autoencoder



$\Rightarrow$
* Linear layer *

Principal Component Analysis



Singular Value Decomposition ($n \geq d$)

$$X = \sum_{i=1}^d \sigma_i u_i v_i^T$$

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_d \geq 0$$

$$v_1, v_2, \dots, v_d \in \mathbb{R}^d$$

$$u_1, u_2, \dots, u_d \in \mathbb{R}^n$$

} orthonormal

EYM Theorem

$$X_k = \underset{\text{rank-}k \tilde{X}}{\operatorname{argmin}} \|X - \tilde{X}\|_F^2$$

$$= \sum_{i=1}^k \sigma_i u_i v_i^T$$

Frobenius and Trace: $\|x\|_F^2 = \text{tr}(x^T x)$

Cyclic Property of Trace: $\text{tr}(AB) = \text{tr}(BA)$

Proof of EYM Theorem: Let $W \in \mathbb{R}^{d \times k}$ have orthonormal cols
 $\Leftrightarrow W^T W = I$

$$\text{rank-}k \quad \tilde{X} = X W W^T$$

$$\|x - \tilde{X}\|_F^2 = \|x - x W W^T\|_F^2 = \|x (I - W W^T)\|_F^2 = \dots$$

$$= \text{tr}(x^T x) - \text{tr}(x^T x W W^T)$$

$$\Rightarrow \underset{\tilde{X}}{\text{argmin}} \|x - \tilde{X}\|_F^2 = \underset{W}{\text{argmin}} \|x - x W W^T\|_F^2 = \underset{W}{\text{argmax}} \text{tr}(x^T x W W^T)$$

Proof Continued:

$$\text{tr}(X^T X W W^T) = \text{tr}(W^T X^T X W)$$

$$= \text{tr}\left(W^T \sum_{i=1}^d v_i \sigma_i^2 v_i^T W\right)$$

$$= \sum_{i=1}^d \sigma_i^2 \text{tr}(W^T v_i v_i^T W)$$

$$= \sum_{i=1}^d \sigma_i^2 \underbrace{\|v_i^T W\|_F^2}_{z_i}$$

$$\begin{cases} X = \sum_{i=1}^d \sigma_i u_i v_i^T \\ X^T X = \sum_{j=1}^d \sum_{i=1}^d \sigma_j \sigma_i v_j u_j^T u_i v_i^T \\ = \sum_{i=1}^d \sigma_i^2 v_i v_i^T \end{cases}$$

1. $z_i \geq 0$

2. $z_i \leq 1$ since W orthonormal columns and $\|v_i\|_2 = 1$

3. $\sum_{i=1}^d z_i = k$ since $\sum_{i=1}^d \text{tr}(W^T v_i v_i^T W) = \text{tr}\left(W^T \sum_{i=1}^d v_i v_i^T W\right)$
 $= \text{tr}(W^T I W) = \text{tr}(I_k) = k$

Show $\sum_{i=1}^d v_i v_i^T = I$

Proof Continued

$$\operatorname{argmin}_{\tilde{X}} \|X - \tilde{X}\|_F^2 = \operatorname{argmax}_{z_i} \sum_{i=1}^d \sigma_i^2 z_i$$

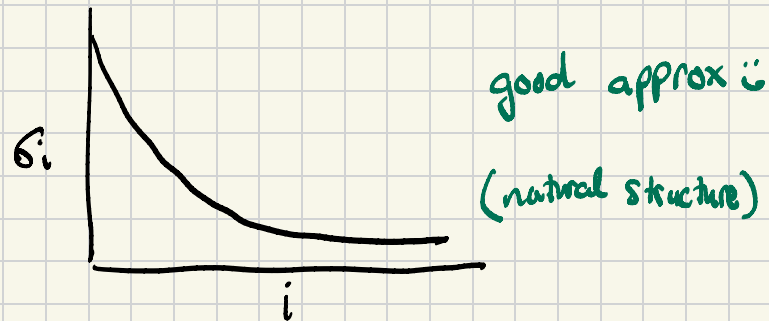
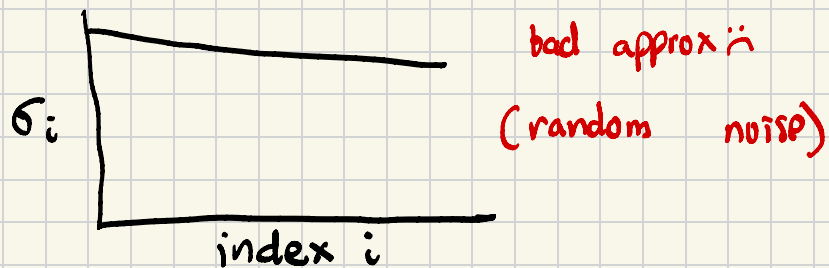
← choose $z_i = \begin{cases} 1 & \text{if } i \leq k \\ 0 & \text{else} \end{cases}$

$$W = \begin{bmatrix} v_1 & v_2 & \dots & v_k \\ | & | & & | \\ 1 & 1 & & 1 \end{bmatrix}$$
$$= \sum_{i=1}^k v_i e_i^T$$

$$\begin{aligned} \tilde{X} &= X W W^T \\ &= \sum_{i=1}^d \sigma_i u_i v_i^T \sum_{j=1}^k v_j e_j^T \sum_{\ell=1}^k e_\ell v_\ell^T \\ &= \sum_{i=1}^k \sigma_i u_i v_i^T \end{aligned}$$

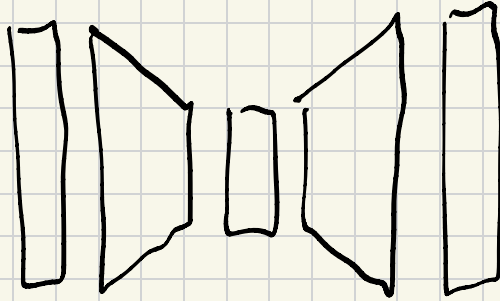
$$\text{Error: } \|X - \tilde{X}\|_F^2 = \dots = \sum_{i=k+1}^d \sigma_i^2$$

Low Rank Approximation



Thursday, April 7

Autoencoder

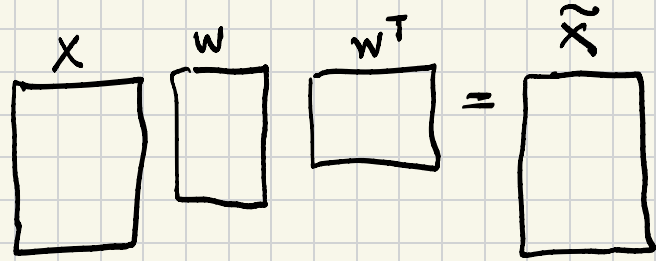


$\Rightarrow$
* linear layers *

PCA

$$X^T W = Z^T$$

$$X^T W W^T = Z^T W^T = \tilde{X}^T$$



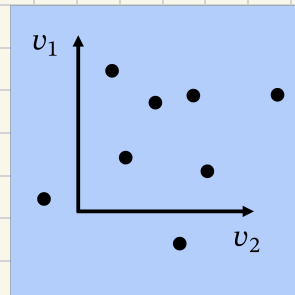
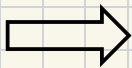
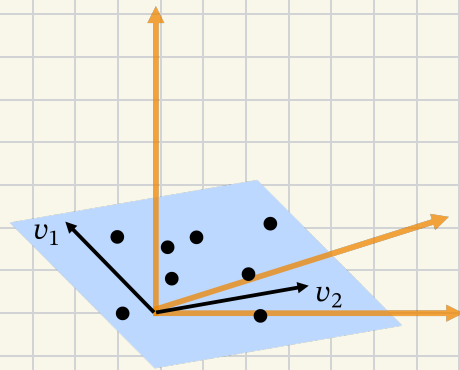
SVD

$$X = \sum_{i=1}^d \sigma_i u_i v_i^T$$

$$X_k = \sum_{i=1}^k \sigma_i u_i v_i^T = \underset{W}{\operatorname{argmin}} \|X - \tilde{X}\|_F^2 = \underset{W}{\operatorname{argmin}} \|X - X W W^T\|_F^2$$

Latent Vector

$$z = f_0(x) = \underset{k \times d}{W}^T \underset{d \times 1}{x} = \sum_{i=1}^k e_i v_i^T x = \begin{bmatrix} \langle v_1, x \rangle \\ \langle v_2, x \rangle \\ \vdots \\ \langle v_k, x \rangle \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_k \end{bmatrix}$$



Reconstruction

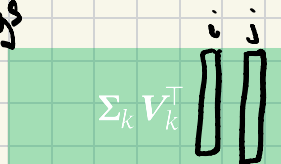
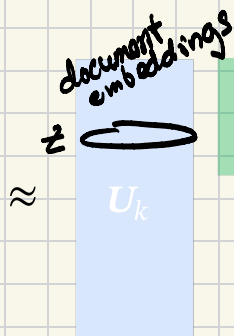
$$f_1(z) = \underset{d \times k}{W}_1 \underset{k \times 1}{z} = \sum_{i=1}^k v_i e_i^T z = \sum_{i=1}^k v_i z_i$$

Semantic Embeddings

Goal: Encode text (+ more) in latent space

Document-word Matrix: n documents, d words

	cat	dog	kitten	puppy	great	good	...
doc ₁			1				1
doc ₂	1				1		1
doc ₃		1			1		1
			1		1		
	1						1
				1			
					1		
						1	
							1

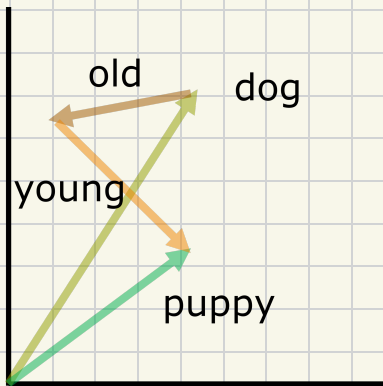
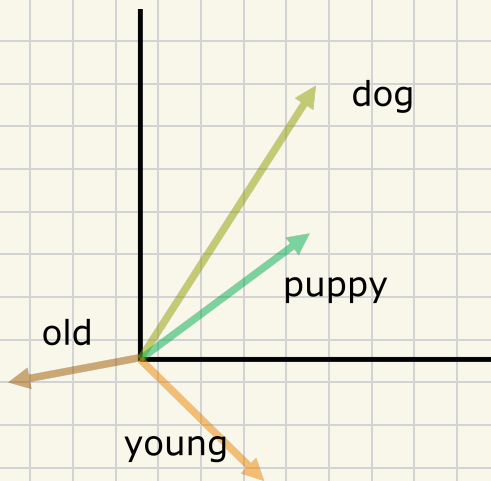


y_i, y_j word embeddings

If $X \approx \tilde{X}$ then:

- If [word i in doc z] = $[X]_{z,i} \approx [\tilde{X}]_{z,i} = \langle z, y_i \rangle \Rightarrow y_i, y_j$ if words i, j in the same docs.
- If [word j in doc z] = $[X]_{z,j} \approx [\tilde{X}]_{z,j} = \langle z, y_j \rangle$

=> Word Math



Similarity Matrix

$$M \in \mathbb{R}^{d \times d}$$

$$\text{Find } Y \in \mathbb{R}^{d \times k} \text{ st } M \approx Y^T Y$$

Let $y_i \in \mathbb{R}^k$ be embedding of i

One Approach: Co-occurrences in text

"I know some good games we could play," said the cat.

"I know some new tricks," said the Cat in the Hat.

"A lot of good tricks. I will show them to you.

Your mother will not mind at all if I do."

	cat	said	good	trick
cat	2	2	0	1
said	2	2	0	1
good	0	0	2	1
trick	1	1	1	2

Challenge: Zipf's Law

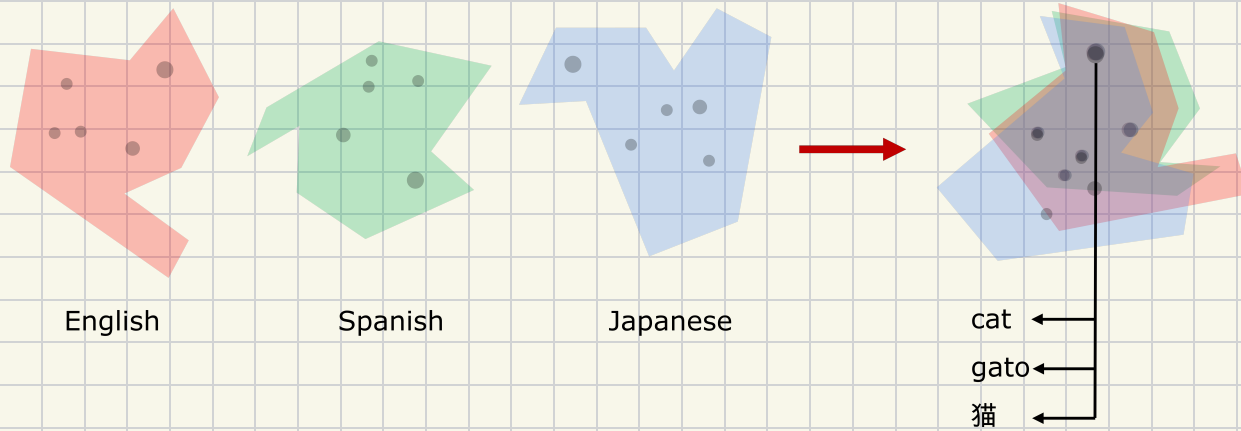
i.e. some words are more common

$$[M]_{i,j} = \# \text{ context w/ } i, j$$

vs

$$[M]_{i,j} = \log \frac{\# \text{ context w/ } i, j}{\# \text{ context w/ } i \cdot \# \text{ w/ } j}$$

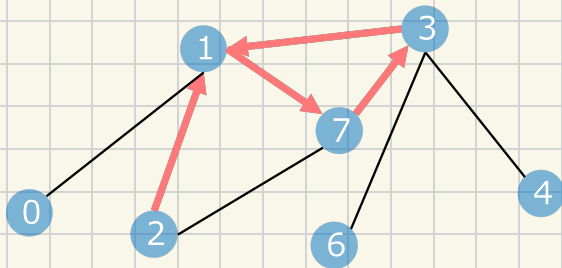
Unsupervised Translation



For each word, compute "fingerprint" of distances and match

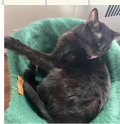
⇒ replaced by LLMs!

Graph Data e.g., social network, internet, etc



	1	2	3	7
1	0	0	0	1
2	1	0	0	0
3	1	0	0	0
7	0	0	1	0

(Multi-modal) Contrastive Learning



Stripes on the grass.	1	0	0	0
Caleb cleaning himself.	0	1	0	0
Stripes stretching out.	0	0	1	0
Caleb posing with Lego.	0	0	0	1

$M = I$ and choose $M \approx W^T Y$,

Equivalently,

$$\text{Loss} = \sum_{i,j} (w_i^T y_j - \mathbb{1}[i=j])^2$$